

Base-16

①

Count to 50



	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
10	10	11	12	13	14	15	16	17	18	19	1A	1B	1C	1D	1E	1F
20	20	21	22	23	24	25	26	27	28	29	2A	2B	2C	2D	2E	2F
30	30	31														
40																
50																

Q1) Smallest Possible Base $\rightarrow 2$ (1, 0)

Q2) Base 10, how Many Numbers with exact 4 digits?

<u>FOR</u>	<u>2 digits</u>	<u>3 digits</u>	<u>4 digits</u>	<u>K digits</u>
	0-99	0-999	0-9999	
	100 Numbers	1000 Numbers	10,000	
	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
	10^2	10^3	10^4	10^K

<u>FOR</u>	<u>Base 16</u>	<u>2 digits</u>	<u>3 digits</u>	<u>4 digits</u>	<u>d digits</u>
		16^2	16^3	16^4	16^d
		$\downarrow$	$\downarrow$	$\downarrow$	
		256	4096	65,536	16^{**d}

$\rightarrow$ Base ** digits

Q3) Base 10, how Many Numbers with d digits?

→ Base * * digits

→ 10 * * d

Q4) Base 7, d digits → 7 * * d

Q5) Base b, d digits → b * * d

		0	1	2	3	4	5	6	7	8	9	¹⁰ A	¹¹ B	¹² C	¹³ D	¹⁴ E
Single Digit Addition Base 15	0	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E
	1	1	2	3	4	5	6	7	8	9	A	B	C	D	E	10
	2	2	3	4	5	6	7	8	9	A	B	C	D	E	10	11
	3	3	4	5	6	7	8	9	A	B	C	D	E	10	11	12
	4	4	5	6	7	8	9	A	B	C	D	E	10	11	12	13
	5	5	6	7	8	9	A	B	C	D	E	10	11	12	13	14
	6	6	7	8	9	A	B	C	D	E	10	11	12	13	14	15
	7															16
	8															17
	9															18
	¹⁰ A							19								19
	¹¹ B															1A
	¹² C															1B
	¹³ D															1C
	¹⁴ E															1D
	E	E	10	11	12	13	14	15	16	17	18	19	1A	1B	1C	1D

Addition

Base-15

(i)
$$\begin{array}{r} 11 \\ + E \\ \hline \end{array}$$

$$\begin{array}{r} \rightarrow 11 \\ 14 \\ \hline 15 \\ \downarrow \\ 10 \\ \downarrow \\ \hline 20 \end{array}$$

(ii)
$$\begin{array}{r} EE \\ 10 \\ \hline \end{array}$$

$\rightarrow$

$$\begin{array}{r} 14 \ 14 \\ 1 \ 0 \\ \hline 15 \ 14 \\ \downarrow \ \downarrow \\ \hline 10 \ E \end{array}$$

(iii)
$$\begin{array}{r} 32 \\ 2E \\ \hline \end{array}$$

$\rightarrow$

$$\begin{array}{r} 32 \\ 2 \ 14 \\ \hline 5 \ 16 \\ \downarrow \ \downarrow \\ 5 \ 1 \\ \hline 6 \ 1 \end{array}$$

(iv)
$$\begin{array}{r} 110 \\ EE \\ \hline \end{array}$$

$$\begin{array}{r} \downarrow E \\ 15 \\ \downarrow \ \downarrow \\ 10 \ E \\ \hline 20 \ E \end{array}$$

(v)
$$\begin{array}{r} 201 \\ 10E \\ \hline \end{array}$$

$$\begin{array}{r} 15 \\ \downarrow \\ 310 \end{array}$$

(vi)
$$\begin{array}{r} 600 \\ 477 \\ \hline \end{array}$$

$$\begin{array}{r} 10 \ 7 \ 7 \\ \downarrow \ \downarrow \ \downarrow \\ \hline A \ 7 \ 7 \end{array}$$

Base-15

Substraction

$$\textcircled{i} \begin{array}{r} 1 \\ -0 \\ \hline 1 \end{array}$$

$$\textcircled{ii} \begin{array}{r} 5 \\ -3 \\ \hline 2 \end{array}$$

$$\textcircled{iii} \begin{array}{r} 9 \\ -3 \\ \hline 6 \end{array}$$

$$\textcircled{iv} \begin{array}{r} A \\ -7 \rightarrow 10 \\ \hline 3 \end{array}$$

$$\textcircled{v} \begin{array}{r} B \\ -A \rightarrow 11 \\ \hline 1 \end{array}$$

$$\textcircled{vi} \begin{array}{r} D \\ -0 \rightarrow 13 \\ \hline 13 \rightarrow D \end{array}$$

$$\textcircled{vii} \begin{array}{r} E \\ -0 \\ \hline E \end{array}$$

$$\textcircled{viii} \begin{array}{r} 7 \\ -5 \\ \hline 2 \end{array}$$

Multi Digit Substraction

$$\textcircled{i} \begin{array}{r} 20^{15} \\ -11 \\ \hline 14 \rightarrow E \end{array}$$

$$\textcircled{ii} \begin{array}{r} EE \\ -1E \\ \hline D0 \end{array}$$

$$\textcircled{iii} \begin{array}{r} 200^{15} \\ -E0 \\ \hline 110 \end{array}$$

$$\textcircled{iv} \begin{array}{r} 110^{15} \\ -EE \\ \hline 11 \end{array}$$

$$\textcircled{v} \begin{array}{r} EEE \\ -110 \\ \hline DDE \end{array}$$

$$\textcircled{vi} \begin{array}{r} 1000^{14} \\ -EE \\ \hline E01 \end{array} \rightarrow 15 \times 1 + 0 + 0 + 0 \rightarrow 3375$$

$$(14 \times 15) + 14 \rightarrow 224$$

$$\underline{3151}$$

$$\begin{array}{r} 15 \overline{) 3151} (210 \\ \underline{3150} \\ 1 \end{array}$$

$$\begin{array}{r} 15 \overline{) 210} (14 \\ \underline{210} \\ 0 \end{array}$$

EOI

$$\begin{array}{ccccccc} & 14 & 14 & 15 & & & \\ 1 & 0 & 0 & 0 & & & \\ & -E & E & & & & \rightarrow \end{array}$$

$$\begin{array}{r} 0 \quad 14 \quad 14 \quad 15 \\ - \quad 14 \quad 14 \\ \hline 14 \quad 0 \quad 1 \\ \hline \downarrow \\ \hline E \quad 0 \quad 1 \\ \hline \hline \end{array}$$

Base to
(Decimal

$$EE \rightarrow (14 \times 15') + (14 \times 15'')$$

$$210 + 14 = 224$$

$$\begin{array}{r} 1000 \\ - EE \\ \hline \end{array} \rightarrow \begin{array}{r} 3375 \\ - 224 \\ \hline \end{array}$$

$$\begin{array}{r} 15 \overline{) 3151} \quad (210 \\ \underline{3150} \\ 1 \end{array}$$

14 0 1

E 01

Decimal To Base

Multiplication

i

$$\begin{array}{r} 11 \\ \times E \\ \hline EE \\ \hline \downarrow \end{array}$$

$$\rightarrow 15 + 1 \rightarrow 16$$

$$\times 14$$

$$\begin{array}{r} 15 \overline{) 224} \quad (14 \\ \underline{210} \\ 14 \end{array}$$

$$\downarrow \quad \downarrow$$

$$14 \quad 14$$

$$\underline{EE}$$

$$= 14 \times 15 + 14$$

$$= 210 + 14$$

$$= 224$$

ii

$$\begin{array}{r} 12 \\ \times 11 \\ \hline 12 \\ 120 \\ \hline 132 \end{array}$$

$$\rightarrow 15 + 2 = 17$$

$$15 + 1 = 16$$

$$\begin{array}{r} \times 16 \\ \underline{102} \\ 17 \\ \underline{272} \end{array}$$

$$\begin{array}{r} 15 \overline{) 272} \quad (18 \rightarrow 18 \quad (\\ \underline{270} \\ 2 \\ \hline \end{array} \quad \begin{array}{r} 18 \\ \underline{15} \\ 3 \end{array}$$

$$\underline{132}$$

iii)

21

$\times E$

$\rightarrow$

21

14

14

$\rightarrow$

21

14

E

$\rightarrow$

21

14

(28) E

$\downarrow$

15 $\overline{) 28} (1$

15

13

$\downarrow$

13

$\downarrow$

10

21

14

10E

iv)

2 E

$\times 12$

$\rightarrow$

2 14

$\times 12$

168

$\rightarrow 15 \overline{) 168} (11$

15

3

$\downarrow$

11 3

$\downarrow$

B 3

2 14

$\times 12$

24 3

(11) $\leftrightarrow$ 8
15 $\overline{) 35} (2$

30

5

$\downarrow$ 25

(iv)

$$\begin{array}{r} 2E \\ \times 12 \\ \hline \end{array}$$

$$\begin{array}{r} 34D \\ \hline \end{array}$$

$$\begin{aligned} &\rightarrow 2 \times 15^1 + 14 \times 15^0 = 30 + 14 = 44 \\ &1 \times 15^1 + 2 \times 15^0 = 15 + 2 = 17 \end{aligned}$$

$$\begin{array}{r} 748 \\ \hline \end{array}$$

$$\begin{array}{r} 15 \overline{) 748} \quad (49 \\ \underline{735} \\ (13 \end{array}$$

$$\begin{array}{r} 15 \overline{) 49} \quad (3 \\ \underline{45} \\ (4 \end{array}$$

$$\begin{array}{r} 15 \overline{) 3} \quad (0 \\ \underline{0} \\ (3 \end{array}$$

$$\begin{array}{r} 3 \quad 4 \quad 13 \\ \hline 3 \quad 4 \quad 0 \\ \hline \end{array}$$

⑤

2 E

$$\rightarrow 2 \times 15 + 14 = 44$$

x 11

$$1 \times 15 + 1 = 16$$

31 E

704

$$\begin{array}{r} 15 \overline{) 704} \quad (46 \\ 690 \\ \hline \textcircled{14} \end{array}$$

$$\begin{array}{r} 15 \overline{) 46} \quad (3 \\ 45 \\ \hline \textcircled{1} \end{array}$$

$$\begin{array}{r} 15 \overline{) 30} \quad (0 \\ 0 \\ \hline \textcircled{3} \end{array}$$

~~14~~ 3 1 14

3 | E

(vi)

E E
x E

DE 1

$$\rightarrow 15 \times 14 + 14 = 224$$

$$\begin{array}{r} \times 14 \\ \hline 3136 \\ \hline \end{array}$$

$$\begin{array}{r} 15 \overline{) 3136} \quad (209 \\ \underline{3135} \\ \hline \end{array}$$

$$\begin{array}{r} 15 \overline{) 209} \quad (13 \\ \underline{195} \\ \hline 14 \end{array}$$

$$\begin{array}{r} 15 \overline{) 136} \\ \underline{135} \\ \hline 1 \end{array}$$

13 14

DE